

Civil Engineering

# Calculus and Differential Methods

## Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

### Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$d/dx(ax^n) = anx^{n-1}$$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\int ax^n dx = ax^{n+1}/(n+1) + C$$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$dy/dx + P(x)y = Q(x)$$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\text{Trapezoidal rule: } \int y dx \approx h/2[y_0 + y_n + 2\sum y_i]$$

## Section B: Applied Questions

### Problem Set 1

6. Solve a civil engineering problem involving road profile gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)
7. Solve a civil engineering problem involving drainage-channel area. Show formula, substitution, correct units and a technical interpretation. (10 marks)
8. Solve a civil engineering problem involving settlement-rate models. Show formula, substitution, correct units and a technical interpretation. (10 marks)
9. Solve a civil engineering problem involving root finding. Show formula, substitution, correct units and a technical interpretation. (10 marks)
10. Solve a civil engineering problem involving numerical integration. Show formula, substitution, correct units and a technical interpretation. (10 marks)

**Problem Set 2**

11. Solve a civil engineering problem involving road profile gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)
12. Solve a civil engineering problem involving drainage-channel area. Show formula, substitution, correct units and a technical interpretation. (10 marks)
13. Solve a civil engineering problem involving settlement-rate models. Show formula, substitution, correct units and a technical interpretation. (10 marks)
14. Solve a civil engineering problem involving root finding. Show formula, substitution, correct units and a technical interpretation. (10 marks)
15. Solve a civil engineering problem involving numerical integration. Show formula, substitution, correct units and a technical interpretation. (10 marks)

**Problem Set 3**

16. Solve a civil engineering problem involving road profile gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)
17. Solve a civil engineering problem involving drainage-channel area. Show formula, substitution, correct units and a technical interpretation. (10 marks)
18. Solve a civil engineering problem involving settlement-rate models. Show formula, substitution, correct units and a technical interpretation. (10 marks)
19. Solve a civil engineering problem involving root finding. Show formula, substitution, correct units and a technical interpretation. (10 marks)
20. Solve a civil engineering problem involving numerical integration. Show formula, substitution, correct units and a technical interpretation. (10 marks)

**Problem Set 4**

21. Solve a civil engineering problem involving road profile gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)
22. Solve a civil engineering problem involving drainage-channel area. Show formula, substitution, correct units and a technical interpretation. (10 marks)
23. Solve a civil engineering problem involving settlement-rate models. Show formula, substitution, correct units and a technical interpretation. (10 marks)
24. Solve a civil engineering problem involving root finding. Show formula, substitution, correct units and a technical interpretation. (10 marks)
25. Solve a civil engineering problem involving numerical integration. Show formula, substitution, correct units and a technical interpretation. (10 marks)

## References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.
2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.
3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.
4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.